Probabilistic Computing for Real-Time, Operationally Cohesive En-route Deconfliction

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***Abstract*—The en-route air traffic deconfliction problem is notoriously difficult, requiring safety-critical and real-time management of highly congested airspaces – in short, a multi-agent control problem with significant operational complications. In this work, we present a novel en-route deconfliction procedure alongside a novel implementation with a probabilistic processing unit. Our approach reformulates the en-route deconfliction problem as a sequential insertion problem whose aim is to maintain operational smoothness. That is, we balance the needs for simple deviations (whose complexity we explicitly quantify), robustness of arrival schedules, perturbing active flights as infrequently as possible, and adhering closely to airlines' preferred trajectories. Concretely, we produce a combinatorial optimisation representation in the form of a quadratic unconstrained binary optimisation (QUBO) problem for safely and robustly traversing a set of interacting graphs. Following implementation on Quantum Dice's ORBIT Probabilistic Processor, we demonstrate a capability for real-time decision-making with desirable performance on synthetic examples of North Atlantic (NAT) airspace conflicts.**

***Keywords*—en-route deconfliction; robust scheduling; probabilistic computing; combinatorial optimisation; quadratic unconstrainted binary optimisation (QUBO) problems**

I. Introduction

Even formulating the en-route air traffic deconfliction problem as a tractable and operationally realistic optimisation problem is challenging. Indeed, the particular sequences of decision-making across all aspects of air travel give rise to complex airline objectives, entangled across different actors with differing sets of information and capacity. It is therefore unsurprising that much of conflict detection and resolution – needing to be prompt and fault-intolerant – is somewhat myopic, prioritising safety when conflicts arise. This potentially defers resolution complexity later into flights, and essentially ignores changes to operational costs [1].

What precipitates our formulation is the observation that operational costs can be understood to comprise flight-local costs, such as individual delays and additional fuel consumption, and global costs, such as changes in schedule. Airports prepare for arrivals and departures carefully, with limited flexibility – changes in schedule propagate throughout the entire network of air traffic. Ultimately, these departures from planned schedules force controllers to engage in airborne-holding manoeuvres, such as holding stacks, resulting in significant increases in fuel consumption for the flights concerned [2], [3]. To this end, our approach relies on the following observations:

- Other than safety, the highest operational priority for deconfliction is to preserve the scheduling of the global departure-arrival network. This will ensure flights do not experience shocks in fuel consumption due to heavy controller intervention arising from airport capacity bottlenecks.
- Within the resulting "operationally smooth" solution envelope, delays and additional fuel consumption will be kept within an interval that is predictable by the airline. This manages the fact that controllers cannot directly optimise the private objectives of airlines.

In general, the conflict resolution problem poses a high-dimensional, self-interacting, combinatorial optimisation problem. Given their ability to model paired interactions, quadratic unconstrained binary optimisation (QUBO) problems serve as a natural candidate (see [4] for an overview), where the optimiser wishes to minimise the objective $H(\boldsymbol{\eta}) := \boldsymbol{\eta}^\top Q \boldsymbol{\eta}$, where $\boldsymbol{\eta}$ denotes a binary decision vector, H denotes the energy function, and Q denotes the (symmetric or triangular) QUBO matrix, which encodes interactions between decisions in the vector $\boldsymbol{\eta}$. Our chosen approach for solving QUBO instances in this work is via Quantum Dice's probabilistic processing unit, based on networks of probabilistic bits (p-bits), which are binary stochastic neurons whose states fluctuate according to controlled probability inputs [5], [6].

For a given inverse temperature β, p-bits evolve via Gibbs sampling, with groups of independent p-bits being updated sequentially [5], [6]. During each update the controlled probability of a given p-bit η_i is set according to its local energy gradient, and the state is then updated via a sigmoid update rule. After a relaxation time, samples taken from the system approximate the distribution $\pi_\beta(\boldsymbol{\eta}) = \frac{1}{Z_\beta} \exp\{-\beta H(\boldsymbol{\eta})\}$, where Z_β is the partition function (a normalising constant). As $\beta \to 0$, the state space is sampled near uniformly, as the high system temperature dominates the exploration dynamics. Conversely, as $\beta \to \infty$, the exploration dynamics become that of deterministic greedy minimisation on account of the low system temperature. Annealing strategies, where system temperature is gradually decreased during execution, are used to find low energy states while escaping poor local minima [7]. For this deconfliction problem, low energy states correspond to configurations satisfying the strongly penalised QUBO constraints and achieving lower intervention complexity.

Two particularly notable works using QUBO formulations

are [8], [9], which attempt to use quantum annealing to address deconfliction. The former uses ground-holding to attempt deconfliction of genuine airspaces, whereas the latter considers benchmark multi-agent pathfinding problems over common spatiotemporal grids. While the results are impressive, the action sets are overly simplistic, fail to address operational smoothness, and do not scale reliably in the number of agents. What is particularly forgiving about bridging to the operational reality is the fact that the fleet need not be constantly and simultaneously controlled. In the operationally fair setting of *first-scheduled, first-served* (as discussed in [10], [11]), we can forego the need for fleet-level control, and instead pursue the required complexity increase for the action sets corresponding to real controller manoeuvres.

Our contributions are as follows:

- We present an operationally smooth reformulation of the conflict detection and resolution problem. This approach avoids offloading operational costs onto auxiliary operational processes;
- Motivated by cases where single-flight insertion is not realisable by a reasonable individual manoeuvre, we explicitly quantify deviation complexity, and present the primary objective as a complexity regularisation term;
- Using efficient information propagation algorithms and disjunctive programming techniques, we are able to map the higher-order problem to a QUBO problem. Our subsequent numerical experiments on the North Atlantic (NAT) airspace, implemented on Quantum Dice's ORBIT Probabilistic Processor (see [12]), demonstrate the ability for real-time decision-making that is consistently safe, simple, and operationally robust.

II. Methodology

A. Formulating Deviations and Conflicts

We consider the problem of optimally inserting flight N into a system of $N-1$ deconflicted flights, which reflects the "always-on" setting of real air traffic control. Importantly, by deconflicting over the full trajectory rather than through local interventions, our formulation can account for the downstream operational consequences of a resolution. Each flight i wishes to follow a nominal trajectory (considered optimal for its airline's private cost function) given by

$$\xi^{(i)} := (\phi_t^{(i)}, \lambda_t^{(i)}, h_t^{(i)})_{t=\tau_0^{(i)}}^{\tau_1^{(i)}} \in [-\frac{\pi}{2}, \frac{\pi}{2}] \times [-\pi, \pi] \times \mathcal{H}_{\text{legal}},$$

which is a time-discretised path (in longitude-latitude-altitude coordinates along waypoints) from position $\xi^{(i)}_{\tau_0^{(i)}}$ at departure time $t = \tau_0^{(i)}$, all the way through to the arrival point $\xi^{(i)}_{\tau_1^{(i)}}$, scheduled for time $\tau_1^{(i)}$. Note also the set $\mathcal{H}_{\text{legal}}$, which represents the levels permitted by controllers – this is a finite, discrete set.

The airline also specifies a speed profile $(m_t^{(i)})_t$, which we leave unchanged. Speed regulation has been extensively studied as a conflict-resolution mechanism [13], [14]. However, speed interventions modify downstream trajectory timing and therefore introduce temporal coupling beyond the immediate conflict being resolved. This has motivated localised or finite-horizon solution strategies in several formulations, which cannot hope to preserve operational smoothness, and still scale poorly. We therefore retain the prescribed aircraft speed profile, allowing weather-dependent traversal times to be precomputed, and consider lateral and vertical interventions for tactical deconfliction.

Indeed, suppose that we re-index each nominal trajectory to $(\xi^{(i)})_{\tau=0}^{T^{(i)}}$, where $T^{(i)}+1$ denotes the number of waypoints for flight i. Associated to each waypoint, we enumerate a flight-local trajectory deviation skeleton. The chosen lateral deviations are detailed by $\{\theta_\tau^{(i)}\}_{\tau=0}^{T^{(i)}}$

$$\theta_\tau^{(i)} \in \{-K, -K+1, \dots, K-1, K\},$$

where $\theta_\tau^{(i)} = k$ denotes a lateral displacement of kd for some pre-specified distance d, relative to the nominal flat profile at waypoint τ for flight i, and K represents some maximal deviation index that we consider. We fix $\theta_0^{(i)}, \theta_{T^{(i)}}^{(i)} = 0$ for departure and arrival.

Similarly, the controller can choose vertical deviations in relation to the chosen vertical profile $\{h_\tau^{(i)}\}_{\tau=0}^{T^{(i)}}$, denoted by

$$\chi_\tau^{(i)} \in \{-J, -J+1, \dots, J-1, J\},$$

where $\chi_\tau^{(i)} = j$ denotes a new level of $(h_\tau^{(i)} + j)$ for flight i at waypoint τ, and we fix $\chi_0^{(i)}, \chi_{T^{(i)}}^{(i)} = 0$. In our framework, we consider the level changes for waypoint $\tau+1$ to be carried out immediately after reaching waypoint τ, though this modelling choice can be changed arbitrarily. Note that the original nominal trajectory $\xi^{(i)}$ can be recovered by selecting $\theta_\tau^{(i)}, \chi_\tau^{(i)} = 0$ for all τ.

For perturbing flight N, we enumerate a trajectory-local graph, whose edges are characterised by the deviation instructions discussed above. Concretely, for perturbable flight N, we define a spatiotemporal undirected graph $G_N := (V_N, E_N)$, where $V_N := \{0, 1, \dots, T^{(N)}\} \times V, E_N := \{0, 1, \dots, T^{(N)}\} \times E$ denote the time-enriched versions of the static vertex and edge options, V, E. Vertices $v \in V_N$ of the full graph are of the form

$$v = (\tau, \theta, \chi),$$

and edges have the feedforward form

$$e = (v, v') = \Big((\tau-1, \theta, \chi), (\tau, \theta', \chi')\Big) \in V_N \times V_N,$$

where $(\theta, \chi) \in \{-K, -K+1, \dots, K-1, K\} \times \{-J, -J+1, \dots, J-1, J\}$ indicate lateral deviation θ and flight level deviation χ at waypoint $\tau - 1$. Since edges have feedforward form, we can rewrite e using just one waypoint index τ, placing it in E_N. Note that, for a non-perturbable flight $j \neq N$, we denote its graph as $G_j^{(\xi)} := (V_j^{(\xi)}, E_j^{(\xi)})$, where vertices are of the form $v = (\tau, 0, 0)$.

Returning to our action space above, the aim of the controller is to leverage these actions at each waypoint to optimally navigate from the starting vertex $v_0^{(N)} = (0, 0, 0)$

to $v_{T^{(N)}} = (T^{(N)}, 0, 0)$, given some objective and constraints that we specify later, via some intermediate path $((\tau, \theta_\tau^{(N)}, \chi_\tau^{(N)}))_{\tau=1}^{T^{(N)}-1}$. Though time-optimality will play a role, it will not form the primary objective of our framework – it is in this sense that our formulation is not a standard shortest-path problem. Nevertheless, we still associate to each edge a cost corresponding to the time to traverse the space. Since the graph is feedforward in waypoint-time, we denote these costs as $c_N(\tau, \theta_{\tau-1}^{(N)}, \chi_{\tau-1}^{(N)}, \theta_\tau^{(N)}, \chi_\tau^{(N)})$, indicating the time it takes flight N to traverse from lateral-vertical deviation pair $(\theta_{\tau-1}^{(N)}, \chi_{\tau-1}^{(N)})$ at waypoint $\tau - 1$ to $(\theta_\tau^{(N)}, \chi_\tau^{(N)})$ at waypoint τ. For $j \neq N$, costs are of the form $c_j(\tau)$.

With our understanding of the deviation skeleton, we can proceed to formulate the conflict sets which underpin the safety constraints in our final combinatorial optimisation problem. In [8] it was simple to do this in a spatiotemporal fashion, as the action set involved explicit delays on a bounded discrete set. In our setting however, where the c_N depend on wind, it is not so clear to biject our set of actions to such a set of delays. Nonetheless, we can define the set of potential conflicts via spatial information, which is fixed beforehand.

We begin by reformulating the trajectory process of flights to that of sequentially occupying subintervals along a chosen trajectory, which are occupied for intervals of time given by the respective edge costs. This not only simplifies the formulation, but also increases robustness. On the other hand, waypoint-to-waypoint subintervals could be too large, so we split every waypoint gap into δ subintervals, so that a trajectory $\xi^{(i)}$ gets split into $T^{(i)}\delta$ subintervals. For notational brevity, we proceed without the additional δ notation, though we will specify our δ setting in Section III.

By definition of our framework, flights in $[N-1] := \{1, \dots, N-1\}$ are deconflicted with one another – the only potential conflicts exist between flight N and those in $[N-1]$. Hence we define spatial subinterval conflict set as

$$C^{(N)} := \bigcup_{j=1}^{N-1} C_j^{(N)},$$

where each $C_j^{(N)}$ denotes a set of pairs of edges, with one edge in deviation structure of flight N, and the other in the nominal trajectory of flight j, that are spatially conflicting. In more detail, we have the characterisation

$$C_j^{(N)} := \{(u, v) \in E_N \times E_j^\xi : \underline{\mathrm{d}}(u, v) \leq \alpha\}, \qquad (1)$$

where α denotes the separation distance standards and $\underline{\mathrm{d}}$ denotes the minimum distance between two edges.

For the incoming flight's trajectory skeleton E_N (or E_j^ξ if $j \neq N$), and chosen deviation instructions $\{(\theta_\tau^{(N)}, \chi_\tau^{(N)})\}_{\tau=1}^{T^{(N)}-1}$ (or $\{(0,0)\}_{\tau=1}^{T^{(j)}-1}$ if $j \neq N$), we may deduce an edge occupation process $(u_\tau^{(N)})_{\tau=1}^{T^{(N)}} \subset E_N$, with elements

$$u_\tau^{(N)} := \Big(\tau, (\theta_{\tau-1}^{(N)}, \chi_{\tau-1}^{(N)}), (\theta_\tau^{(N)}, \chi_\tau^{(N)})\Big),$$

and a respective post-departure time-of-arrival process $\mathcal{T}^{(N)}$, defined using the partial sum of edge costs

$$\mathcal{T}^{(N)} := \left(\sum_{\gamma=1}^{\tau} c_N(\gamma, \theta_{\gamma-1}^{(N)}, \chi_{\gamma-1}^{(N)}, \theta_\gamma^{(N)}, \chi_\gamma^{(N)})\right)_{\tau=1}^{T^{(N)}},$$

which is defined accordingly for the active flights $j \neq N$,

$$\mathcal{T}^{(j)} := \Big(\sum_{\gamma=1}^{\tau} c_j(\gamma)\Big)_{\tau=1}^{T^{(j)}}.$$

Ideally, we would like to ensure that all edges in spatial conflict are non-overlapping in time. Before making this explicit, we must augment $C^{(N)}$ to account for flight levels. Indeed, we define

$$C_{\mathcal{H}_{\text{legal}}}^{(N)} := \bigcup_{l=0}^{H} C_{h_l}^{(N)},$$

where we make explicit that $\mathcal{H}_{\text{legal}} = \{h_0, h_1, \dots, h_H\}$, and define the spatially conflicting, level-compatible sets $C_{h_l}^{(N)}$ for $h \in \mathcal{H}_{\text{legal}}$ as

$$C_h^{(N)} := \big\{\big(u_\tau^{(N)}, u_{\tilde{\tau}}^{(j)}\big) \in C^{(N)} : \chi_\tau^{(N)} + h_\tau^{(N)} = \chi_{\tilde{\tau}}^{(j)} + h_{\tilde{\tau}}^{(j)} = h\big\},$$

where $u_\tau^{(N)}, u_{\tilde{\tau}}^{(j)}$ are as defined above.

With this notation we may now denote the time occupation interval associated to general flight i from waypoint $\tau - 1$ to τ as

$$I_\tau^{(i)} := [\tau_0^{(i)} + \mathcal{T}_{\tau-1}^{(i)} - \frac{\Delta}{2}, \tau_0^{(i)} + \mathcal{T}_\tau^{(i)} + \frac{\Delta}{2}],$$

where we have added Δ to reflect the fact that spatiotemporal deconfliction must include a nonzero buffer window. Now, we may clearly express the deconfliction condition as requiring

$$I_\tau^{(N)} \cap I_{\tilde{\tau}}^{(j)} = \emptyset, \qquad \forall (u_\tau^{(N)}, u_{\tilde{\tau}}^{(j)}) \in C_{\mathcal{H}_{\text{legal}}}^{(N)}. \qquad (2)$$

There are multiple ways in which a controller might satisfy the deconfliction condition (2):

- They may pick edges such that the resulting pairings are spatially deconflicted (so that resulting edges occupy $(C^{(N)})^c$);
- They may pick differing flight levels to avoid spatial conflicts (so that the resulting edge pairs might occupy $C^{(N)}$ but not $C_{\mathcal{H}_{\text{legal}}}^{(N)}$;
- They might pick well-timed edges and flight levels to avoid simultaneous occupation of the spatially conflicting, same-level subintervals (so that the resulting edge pairs might occupy $C_{\mathcal{H}_{\text{legal}}}^{(N)}$, but the occupation interval sequences satisfy the Δ-buffered exclusive occupation condition above).

B. Mixed Integer Programming Formulation

As alluded to above, though local trajectory modifications are preferred to global ones, we certainly wish to avoid repeated application of small deviations (inevitable in a myopic setting). Aside from pilots needing simple instructions, this also reflects the fact that highly oscillatory trajectories result in significant fuel expenditure. Though we cannot mitigate this

directly, we can implicitly address it via deviation complexity regularisation.

To this end, we introduce a primary objective of the form

$$\mathcal{C}_N(\boldsymbol{\theta},\boldsymbol{\chi}) := \sum_{\tau=1}^{T^{(N)}} \Big(w_\theta |\theta_\tau^{(N)} - \theta_{\tau-1}^{(N)}|^p + w_\chi |\chi_\tau^{(N)} - \chi_{\tau-1}^{(N)}|^p \Big),$$

where:

- w_θ, w_χ denote controller preferences for the deviation costs. Given the interpretability of the deviation skeleton, this can be easily specified by a controller. For instance, if a controller would always prefer to perform a level change before a lateral deviation, we might set $w_\theta = 2, w_\chi = 1$;
- $p \geq 1$ denotes the type of regularisation chosen. Common examples include lasso-style regularisation with $p = 1$, or ridge-style regularisation with $p = 2$;
- Rather than penalising just the magnitudes of individual deviations, we avoid oscillatory trajectories via the magnitudes of the changes in deviations from waypoint to waypoint. While also accounting for magnitude of oscillation, this objective therefore also penalises the number of oscillations – in other words, the number of deviations.

It is worth noting that, for instance, a constant deviation of $(\theta_\tau^{(N)})_\tau = (1, \dots, 1)$ will be priced the same as a local deviation $(\theta_\tau^{(N)})_\tau = (1, 0, \dots, 0)$. However, in pursuit of operational smoothness, we introduce a scheduling condition which will implicitly differentiate between the two: the former will be deemed unacceptable from a scheduling perspective, and thus rendered infeasible by the constraint.

The scheduling constraint (5) can be found in the *master problem* below. Namely, the controller wishes to minimise the map

$$(\theta_\tau^{(N)}, \chi_\tau^{(N)})_{\tau=1}^{T^{(N)}-1} \mapsto \mathcal{C}_N(\boldsymbol{\theta},\boldsymbol{\chi}), \tag{3}$$

$$\begin{aligned} \text{subject to} \quad & \theta_\tau^{(N)} \in \{-K, -K+1, \dots, K-1, K\}, \\ & \chi_\tau^{(N)} \in \{-J, -J+1, \dots, J-1, J\}, \\ & (\chi_\tau^{(N)} + h_\tau^{(N)}) \in \mathcal{H}_{\text{legal}}, \qquad (4) \\ & |\tau_0^{(N)} + \mathcal{T}_{T^{(N)}} - \tau_1^{(N)}| \leq \mathcal{T}, \qquad (5) \\ & I_\tau^{(N)} \cap I_{\tilde{\tau}}^{(j)} = \emptyset \quad \forall (u_\tau^{(N)}, u_{\tilde{\tau}}^{(j)}) \in C_{\mathcal{H}_{\text{legal}}}^{(N)}, \qquad (6) \end{aligned}$$

where we have introduced the permissible levels set $\mathcal{H}_{\text{legal}}$ in condition (4), and the allowed scheduling perturbation $\mathcal{T}$ in condition (5).

Our novel scheduling-envelope approach is motivated by the following observations:

- As with many operational problems, near-optimal solutions obtained quickly are often more desirable than optimal ones obtained infeasibly slowly. By reformulating the delays as part of an envelope condition, we are able to find operationally feasible solutions quickly.
- A valid approach would view the delays as a primary objective, with the deviation complexity acting as a regulariser. This multi-objective approach suffers from a lack of interpretability, with it being unclear how to compare the unit costs of deviation complexity and arrival delay. We see in our experimental results that the relationship is highly unclear. On the other hand, a maximally acceptable delay is operationally interpretable and therefore easily tuneable.
- Though we do not explore this here, our optimal insertion model will sometimes need to handle more than one flight simultaneously. As the set of perturbable flights grows, retaining individual scheduling envelopes naturally tends towards a fleet-level minimax formulation. This provides greater distributional robustness in schedule deviation by avoiding the potentially unfair concentration of delays on a small set of flights.

C. QUBO Formation

As shown in the deconfliction condition (6), our problem exists at the higher level of edge-selection, rather than vertex selection. Even then, there is paired interaction between edges, since we must guarantee edge selections corresponding to valid paths, and in (6) edges between flights interact. As discussed, these paired interactions naturally lend themselves to QUBO formulation. We lay out the formulation briefly below, providing detail where implementation is non-trivial.

For the conversion of the decision vector $(\boldsymbol{\theta},\boldsymbol{\chi})$ to a binary decision vector, we use one-hot encoding, thus introducing a further constraint into the system. Without the added slack variables relating to the inequalities in the master problem, the initial dimension of the decision vector is $(T^{(i)}-2)(2K+1)^2(2J+1)^2 + 2(2K+1)(2J+1)$ – importantly, the dimensionality scales well with the number of waypoints, and by definition the number of active flights.

The difficult part of our formulation lies in the deconfliction condition (6). Indeed, consider some conflicting edge pair $(u_\tau^{(N)}, u_{\tilde{\tau}}^{(j)}) \in C_{\mathcal{H}_{\text{legal}}}^{(N)}$, with $j \neq N$. If that edge pair is chosen (rather, if the entry in $\boldsymbol{\eta}$ corresponding to edge $u_\tau^{(N)}$ is made active), then we need to enforce either

$$\tau_0^{(N)} + \mathcal{T}_\tau^{(N)} + \Delta \leq \tau_0^{(j)} + \mathcal{T}_{\tilde{\tau}-1}^{(j)} \tag{7}$$

or

$$\tau_0^{(j)} + \mathcal{T}_{\tilde{\tau}}^{(j)} + \Delta \leq \tau_0^{(N)} + \mathcal{T}_{\tau-1}^{(N)}. \tag{8}$$

In other words, we need to either ensure flight N exits the edge Δ-before the conflicting, non-perturbable flight enters its edge, or the converse. For this disjunctive programming problem, we must set a pair of inequalities that can be simultaneously enforced, with the flexibility to handle these cases. We opt for a big-M formulation (see [15], [16] for an overview of such techniques), which simultaneously enforces (7) and (8) via

$$\begin{aligned} \tau_0^{(N)} + \mathcal{T}_\tau^{(N)} + \Delta - (\tau_0^{(j)} + \mathcal{T}_{\tilde{\tau}-1}^{(j)}) \\ \leq M z(u_\tau^{(N)}) + M(1 - \mathbb{I}(u_\tau^{(N)})), \qquad (9) \\ \tau_0^{(j)} + \mathcal{T}_{\tilde{\tau}}^{(j)} + \Delta - (\tau_0^{(N)} + \mathcal{T}_{\tau-1}^{(N)}) \\ \leq M(1 - z(u_\tau^{(N)})) + M(1 - \mathbb{I}(u_\tau^{(N)})), \qquad (10) \end{aligned}$$

where $z(u_\tau^{(N)})$ denotes an additional binary decision variable for the ordering of the two flights, and $\mathbb{I}(u_\tau^{(N)})$ is an indicator function denoting whether the edge $u_\tau^{(N)}$ has actually been selected or not. For instance:

- If edge $u_\tau^{(N)}$ is not even selected, then there is a big-M contribution in both constraints (9), (10), so they are both trivially satisfied;
- In the case that we prefer the incoming flight N to arrive after flight j exits, we choose $z(u_\tau^{(N)}) = 1$, effectively collapsing (10) to (8), and contributing a big-M to (9) to render it trivially enforced.

Upon introducing slack variables to convert inequalities to equalities, squaring, and conversion of all controllable terms to functions of decision vector $\boldsymbol{\eta}$, it is possible to formulate the energy function of the QUBO problem as a quadratic form

$$H(\boldsymbol{\eta}) = \boldsymbol{\eta}^\top Q_N \boldsymbol{\eta},$$

where the QUBO matrix Q_N can be written as

$$Q_N = Q_{\mathcal{P}} + \lambda_{\mathcal{H}} Q_{\mathcal{H}} + \lambda_{\mathcal{T}} Q_{\mathcal{T}} + \lambda_{\mathcal{O}} Q_{\mathcal{O}} + \lambda_{\mathcal{E}} Q_{\mathcal{E}} + \lambda_{\mathcal{M}} Q_{\mathcal{M}},$$

where $Q_{\mathcal{P}}$ denotes the primary cost matrix arising from $C_N(\boldsymbol{\theta}, \boldsymbol{\chi})$, $Q_{\mathcal{H}}$ denotes the legal level enforcement matrix arising from (4), $Q_{\mathcal{T}}$ denotes the matrix associated to scheduling constraint (5), $Q_{\mathcal{O}}$ denotes the matrix enforcing one-hot encoding, $Q_{\mathcal{E}}$ ensures that edge choices link, and $Q_{\mathcal{M}}$ denotes the matrix realised from accounting for the constraints (9), (10). The $\lambda_\cdot$ constants are chosen to balance well-posedness of the energy landscape H and the hardness of the respective constraints. As shown in [8], this hyperparameter choice is often flexible.

D. Probabilistic Computing

Having cast the deconfliction problem in QUBO form $H(\boldsymbol{\eta}) = \boldsymbol{\eta}^\top Q_N \boldsymbol{\eta}$, we now describe the specific hardware used to solve it. Our experiments are carried out on Quantum Dice's ORBIT probabilistic processing unit [12], an opto-electronic implementation of the p-bit network introduced above.

ORBIT pairs a high-speed, certifiable photonic quantum random number generator (QRNG) with an electronic control system. The photonic subsystem employs indium phosphide photonic integrated circuits whose entropy output is certified in real time via a source-device independent protocol [17], guaranteeing the unbiased random inputs on which convergence of the Gibbs sampler depends. Each physical p-bit is time-demultiplexed into multiple logical p-bits by the electronic control unit, which stores the coupling matrix Q and computes the sigmoid threshold for each update. The system is therefore readily reprogrammable to arbitrary QUBO instances without modifying the photonic hardware, with the current system supporting up to 32 000 logical p-bits with arbitrary connectivity or 64 000 with sparse connectivity.

A distinguishing feature of this architecture is its energy profile. The photonic entropy source and electronic control combined consume approximately 1.6 pJ per individual p-bit update, while the overall system draws approximately 20 W for a typical run. This, combined with a compact form factor, makes the platform well suited to the power- and size-constrained edge deployments typical of operational deconfliction scenarios.

III. Numerical Experiments

A. NAT Region Conflict Instances

To avoid choosing between difficult modelling of departure and approach aircraft dynamics (which is arguably a different problem) and clipping these sections of flights (a fairly arbitrary procedure), in this section we present a stylised example of the deconfliction framework introduced.

Taking sample flight paths over the North Atlantic (NAT) airspace (shown in [18]), randomly sampling and subsequently closely sequencing entry times, we are able to produce a set of 63 operationally faithful test instances for the incoming flight deconfliction problem. Each test set consists of 30 Eastbound flights, with the incoming flight conflicting with the currently active flights. For this particular case, it is not necessary to consider both directions, since they either occupy mutually exclusive $\mathcal{H}_{\text{legal}}$ flight-level sets, or are scheduled at different times. Upon construction of these test instances, we are able to categorise the instance type by the severity of confliction with the incoming flight. Categorising the proportion of the incoming flight that is in no conflict, solely a spatial conflict (ignoring levels), and a spatiotemporal conflict (with the Δ-buffer included), we visualise the range of problem types encountered in the simplex of Figure 1. We see that a broad range of test-case types are considered, with the failed trials not necessarily following any particular pattern. Upon further inspection, it is possible to use the dynamic programming principle to compute upper and lower bounds on the exit-arrival gaps on the left in conditions (9), (10). Denoting the largest (resp. smallest) time for flight N to traverse waypoint τ as $\overline{\mathcal{T}}_\tau^{(N)}$ (resp. $\underline{\mathcal{T}}_\tau^{(N)}$), for each of these failed trials we compute the metrics

$$\begin{aligned}
\underline{T}_{N,j}(\tau, \tilde{\tau}) &:= \tau_0^{(N)} + \underline{\mathcal{T}}_\tau^{(N)} + \Delta - (\tau_0^{(j)} + \mathcal{T}_{\tilde{\tau}-1}^{(j)}),\\
\overline{T}_{N,j}(\tau, \tilde{\tau}) &:= \tau_0^{(N)} + \overline{\mathcal{T}}_\tau^{(N)} + \Delta - (\tau_0^{(j)} + \mathcal{T}_{\tilde{\tau}-1}^{(j)}),\\
\underline{T}_{j,N}(\tilde{\tau}, \tau) &:= \tau_0^{(j)} + \mathcal{T}_{\tilde{\tau}}^{(j)} + \Delta - (\tau_0^{(N)} + \overline{\mathcal{T}}_{\tau-1}^{(j)}),\\
\overline{T}_{j,N}(\tilde{\tau}, \tau) &:= \tau_0^{(j)} + \mathcal{T}_{\tilde{\tau}}^{(j)} + \Delta - (\tau_0^{(N)} + \underline{\mathcal{T}}_{\tau-1}^{(j)}),
\end{aligned}$$

corresponding respectively to the smallest and largest possible exit-entry gaps for both orderings (7), (8) of flights N and j. Without constraints, the intervals $[\underline{T}_{N,j}, \overline{T}_{N,j}], [\underline{T}_{j,N}, \overline{T}_{j,N}]$ represent the most temporal separation we can achieve with our deviation skeleton. For the 4 failed test cases we found that spatiotemporal conflict existed at the final waypoint-to-waypoint edge of flight N, where separation can only be achieved temporally, since we cannot deviate from the preferred exit point (the destination in the broader framework). Indeed, temporal separation was found to be impossible, since both minima were positive: $\underline{T}_{N,j}(T^{(N)}, \tilde{\tau}), \underline{T}_{j,N}(\tilde{\tau}, T^{(N)}) > 0$. Rather than reflecting a failure of the optimisation, these therefore reflect an overly restrictive QUBO formulation, since

no solutions exist. The solution set can be made nonempty by increasing the values of K and J, which were admittedly tight in our proof-of-concept experiments. What is encouraging is the fact that – even under such tight manoeuvring allowance – the 4 test instances with no non-conflicting regions and majority spatiotemporal conflict were deconflicted successfully.

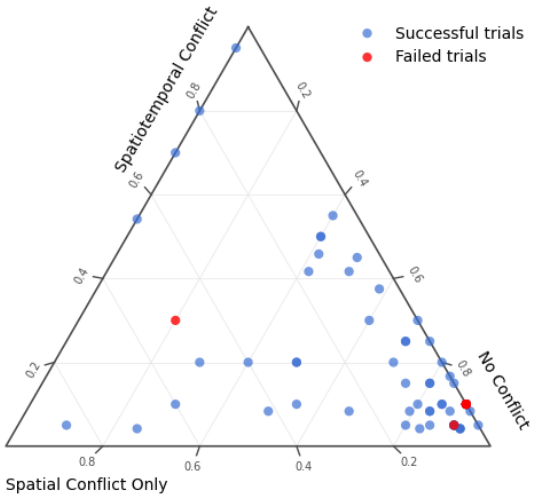


Figure 1. Variation of test instances according to confliction type proportions with incoming flight

B. Optimisation Results

Before discussing results further, we explicitly state our setup:

- We take the permissible Eastbound flight levels to be $\mathcal{H}_{\text{legal}} = \{310, 330, 350, 370, 390, 410\}$, the set of potential deviations is minimal at $J, K = 1$, and we take a scheduling threshold of $\mathcal{T} = 10$;
- For our traversal time computations, we use the ERA5 global climate and weather reanalysis dataset, specifically that from 1st March, 2025 [19];
- Regarding the primary objective $\mathcal{C}_N$, we take the regularisation objective with power $p = 2$, and weights $w_\theta = 2, w_\chi = 1$, to reflect a preference for flight level changes over changes to the flat profile of a flight;
- For the optimisation itself, we perform $200,000$ sweeps in total, starting at an inverse temperature of $\beta = 0.0001$, increasing according to a linear scheduler up to $\beta = 32$;
- For our separation standards, we follow the standard NAT requirements of separation $\alpha = 30$ nautical miles, and a time separation buffer of $\Delta = 3$ minutes.

For each instance, we ran 10 attempts to account for failures due to early termination, reporting the lowest-energy feasible solution, and subsequently the first feasible solution to achieve within 99% of this energy. We also note the runtime to first hit the feasible solution set.

In Figure 2 we display the achieved primary costs for successful optimisations, decomposed according to the lateral and vertical deviation complexities, without weighting by w_θ, w_χ. The final realised primary objective with these weightings is visualised in the surrounding contour plot, which reflects our preference for flight-level changes. The dashed red lines reflect the deviation complexity, in each manoeuvre, arising from minimal intervention – that is, a single flight level change, or a single lateral deviation. Across both the individual trials (in white) and the best per instance (in red), we see that deconfliction is often achieved by such minimal interventions, thus recovering standard, simple interventions while simultaneously guaranteeing robustness of arrival schedule. This adherence to schedule is made explicit in Figure 3,

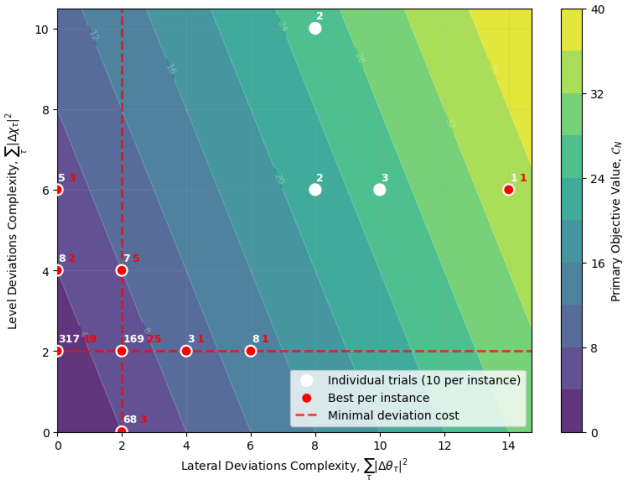


Figure 2. Final achieved deviation complexities. Individual trials (10 per instance) in white, best per instance in red. The red dashed line illustrates minimal deviation in each manoeuvre

where we display the delays achieved by the best solutions per instance from Figure 2, alongside the total intervention complexity. As expected, we generally see that more complex deviation procedures result in larger arrival-time deviations, reflecting the general intuition that simpler interventions will result in better operational smoothness. However, at the level of smaller intervention complexity, such a trend does not seem to exist. This illustrates why an action set parametrised solely by delay may be operationally inadequate – due to the nonlinearities induced by the wind fields, there is no clear bijection from prescribed manoeuvres to delays. In fact, we see that sometimes more complex manoeuvres achieve lower arrival-time deviation. A naive attempt to minimise arrival-time deviation may therefore result in highly involved deviations from the optimal trajectory, producing operationally unacceptable changes, but also potentially trading off extra minutes of fuel consumption for manoeuvres that consume significantly more. It is for these reasons that we choose to minimise deviation complexity $\mathcal{C}_N$ subject to an operationally acceptable schedule flexibility, $\mathcal{T}$. Throughout Figures 2 and 3, we note one particular instance in which the realised primary objective is significant at $\mathcal{C}_N = 34$, with an almost unacceptable schedule deviation of 9.25 minutes. Investigating this case further, we visualise the conflicting flights in Figure 4, colouring cells according to the involved conflicts

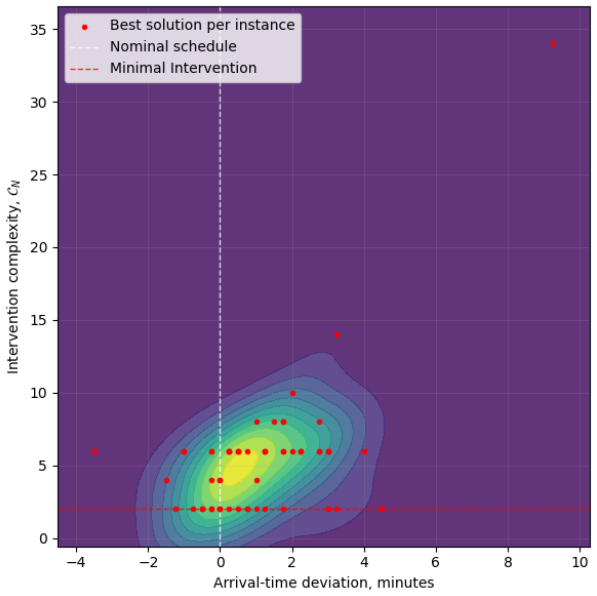


Figure 3. Intervention complexity against arrival-time deviation for best per instance. The white dashed line denotes the nominal scheduling line, and the red dashed line denotes the minimal achievable deviation complexity

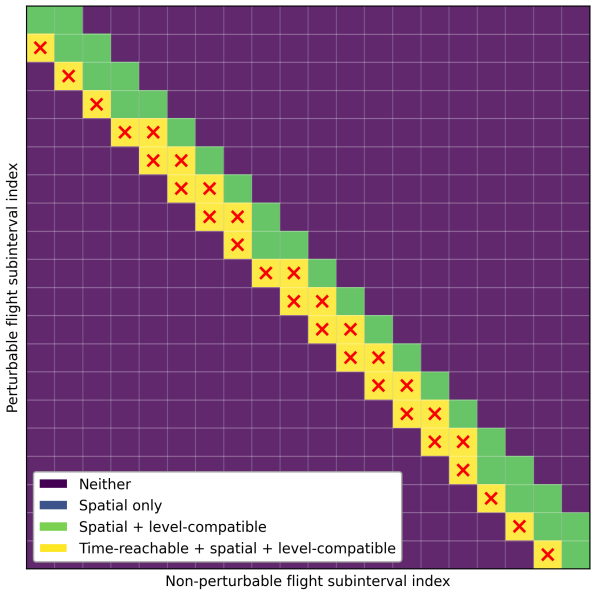


Figure 4. Visualisation of conflict instances across the progression of the incoming flight and its conflicting flight, for the worst-case result

across the progression of each flight. Indeed, we see that the incoming flight follows the exact same trajectory as the conflicting one, shifted backwards by one subinterval – it enters the NAT region only 7.25 minutes after the already active one, which is considerably tight in separation. This particular case demonstrates an ability for our objective to expose the disutility of the highly oscillatory trajectories that result from local intervention strategies. In practice, higher values of the deviation skeleton size K, J would have provided additional non-oscillatory deviations to achieve the separation. Alternatively, in the interest of splitting the required arrival-time deviation across both flights, deviating both may have resulted in a significantly lower total deviation complexity and total arrival-time deviation. Any decision rule for overriding the first-filed, first-served rule would need to be carefully considered. Such a rule, which would act as an operational interpolator from our insertion problem to the full multi-agent fleet control problem, is left for future work, though it would computationally only amount to augmenting the QUBO matrix.

A final important consideration for operational usefulness is the ability for this algorithm to run in real-time, concurrently with live air traffic flow. Only with such rapid decision-making would our insertion problem be viable as an improvement upon local methods. In Figure 5 we visualise this information across the three metrics of time to solution, time to 99% energy optimality, and finally the time to hit the first feasible solution. We observe considerably shorter runtimes than those reported for several exact mathematical-programming approaches to deconfliction (see [13], [20]), with further reductions to be had as the requirement for optimal manoeuvre instructions is relaxed. This impressive time to feasibility highlights a particularly useful benefit of our framework for operational purposes. By formulating the schedule adherence as a constraint envelope, rather than a primary objective to be minimised, we are able to clearly define a notion of weak solution to the problem, allowing us to stop early in cases where runtime is prioritised over full optimality. Such a weak solution is not available in the delay minimisation formulation of the problem.

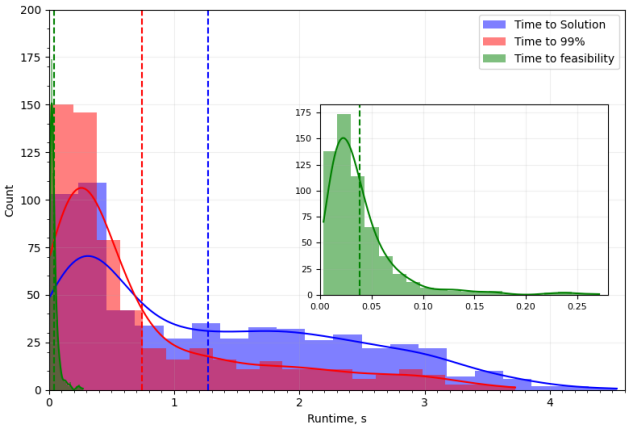


Figure 5. Time to solution for successful trials, alongside time to 99% optimal energy, and time to feasibility

IV. Conclusion

Starting from the operational reality of the *first-filed, first-served* system for air traffic control, we have presented an operationally cohesive reformulation of the optimisation problem relating to real-time en-route deconfliction. The formulation treats operational smoothness as the central objective: intervention complexity is minimised subject to a permissible scheduling envelope, thereby limiting the propagation of

deconfliction-induced disruption through the wider air traffic network. As demonstrated by our experiments on the ORBIT Probabilistic Processor, this effectively reconciles the realities that:

- Small time deviations do not necessarily correspond to simple manoeuvres;
- Minimising a mean delay can lead to unfair individual flight delays in the first instance, and unpredictable shocks to global arrival-departure schedules in the broader context;
- A full-fleet optimisation is operationally and computationally infeasible, but a local intervention strategy (deconflicting locally when conflicts arise) can lead to unforeseen delays and highly complex manoeuvres. Our formulation sits at the intermediate level of optimal insertion, but for full trajectories, and crucially performs quickly enough to facilitate real-time decision-making.

Two extensions are particularly natural. Firstly, although sequential insertion provides a tractable and operationally interpretable baseline, difficult conflicts may be resolved more efficiently by perturbing a small number of active flights and distributing the required deviation between them. A selective multi-flight perturbation rule could therefore interpolate between first-filed, first-served insertion and full fleet-level optimisation, while explicitly accounting for the equitable allocation of disruption between airlines. The difficult instance identified in our experiments provides an example in which such shared intervention may substantially reduce the total required deviation.

Secondly, the fixed symmetric scheduling envelope considered here can be generalised to a dynamic and asymmetric envelope, reflecting that the operational costs of earliness and lateness need not be equal and may change with network state and time of day. Such envelopes could be updated as operational information becomes available, allowing the deconfliction layer to exploit changing schedule flexibility without requiring access to airlines' private objective functions. Probabilistic or learning-based approaches—including Bayesian updating or reinforcement learning—provide possible mechanisms for estimating these state-dependent flexibilities. Together, these extensions point towards a deconfliction architecture that remains locally actionable while becoming increasingly responsive to network-level operational conditions.

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